

Designing, implementing, training, and deploying our own Transformer-based language model from scratch for a specific domain.

That means:

- We write the model architecture.
- We create the tokenizer.
- We build the vocabulary.
- We prepare the dataset.
- We implement the training loop.
- We train our own weights.
- We build the inference engine.
- We integrate RAG.
- We develop the application.

No pretrained LLMs.

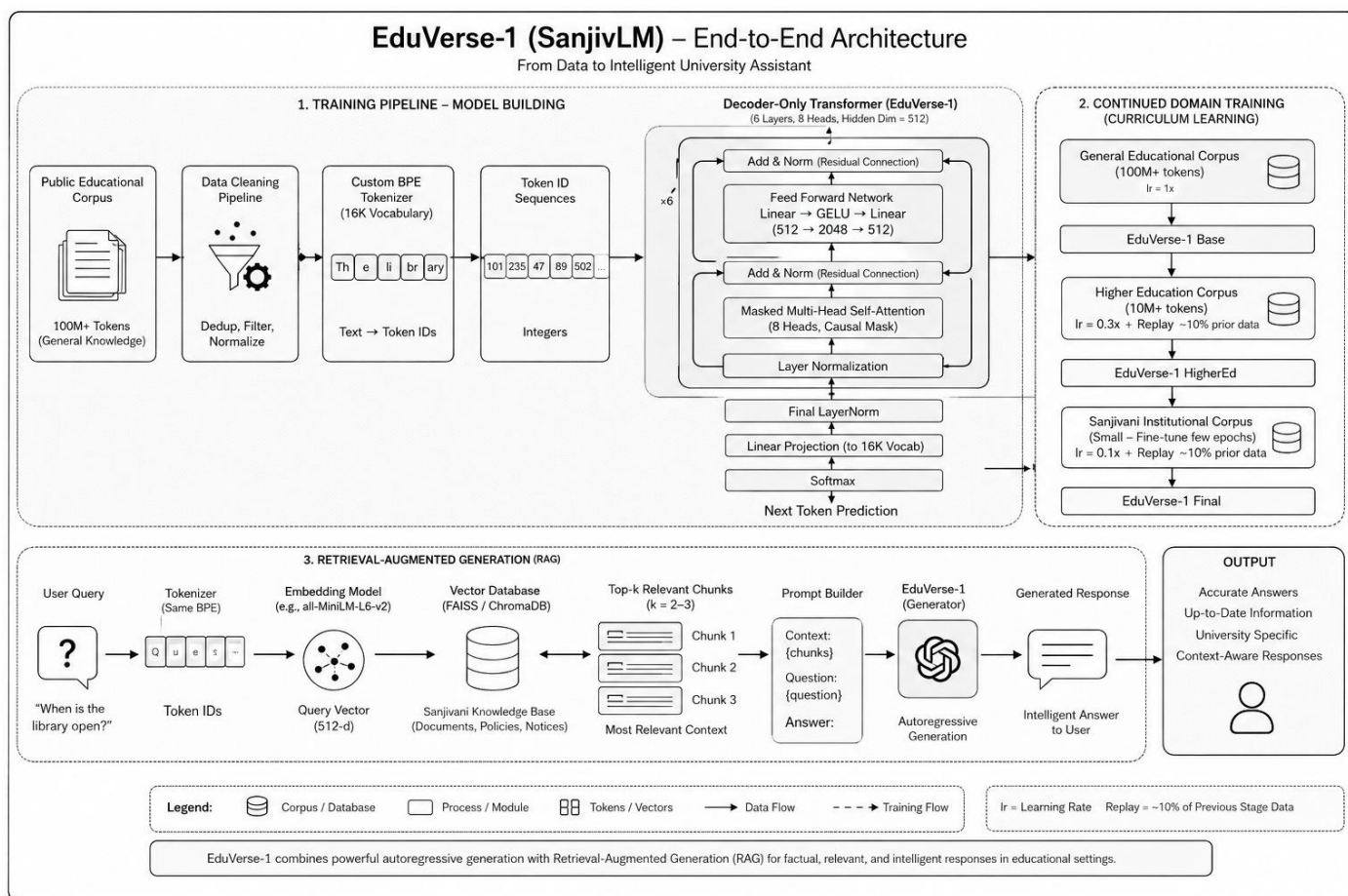
No Frontier API.

EduVerse-1 is a decoder-only Transformer-based Large Language Model developed from scratch for the Higher Education domain. It is trained on publicly available educational resources and further specialized using institutional knowledge from Sanjivani University.

Why Decoder-Only

EduVerse-1 does next-token prediction on one continuous stream of text there's no separate "input language" and "output language" to translate between (unlike the original encoder-decoder Transformer, built for machine translation). A single causal-masked stack is sufficient at each step, the model only attends to tokens that came before it and predicts the next one.

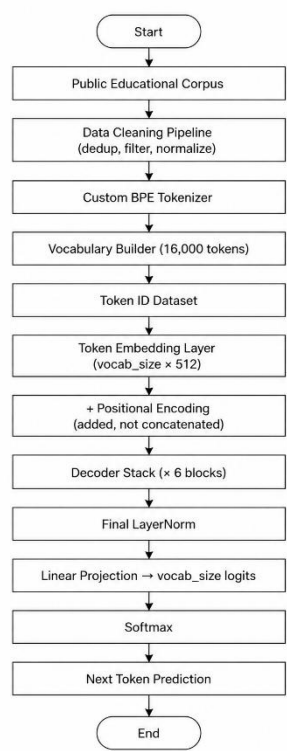
Architecture



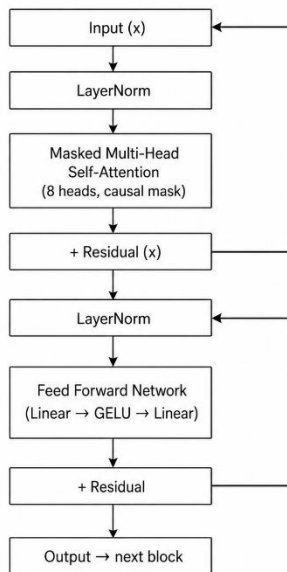
Model Configuration

Component	Value
Architecture	Decoder-only Transformer
Layers	6
Attention Heads	8
Hidden Dimension	512
FFN Inner Dimension	2048 (4× hidden)
Context Length	512 tokens
Vocabulary Size	16,000
Positional Encoding	Learned (or RoPE - see §6)
Normalization	Pre-LayerNorm
Activation	GELU
Parameters	25–35M
Tokenizer	Custom BPE
Framework	PyTorch
Training Objective	Next Token Prediction (cross-entropy)

Full Pipeline (Data → Trained Model)



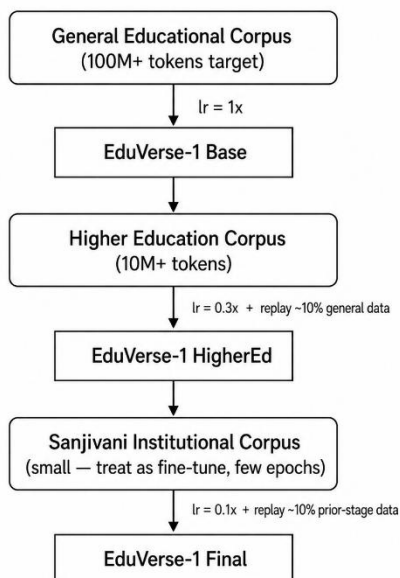
Transformer Decoder Block (Pre-Norm)



Why pre-norm: normalizing *before* attention/FFN (rather than after) keeps gradients stable through many stacked layers important since you won't have infrastructure to babysit a diverging loss curve.

Why causal mask: each position can only attend to itself and earlier positions this is what makes generation autoregressive (no peeking at future tokens).

5. Three-Stage Training Curriculum



Lower learning rate and data replay at each later stage guard against catastrophic forgetting without this, each fine-tuning stage risks overwriting general language ability learned earlier.

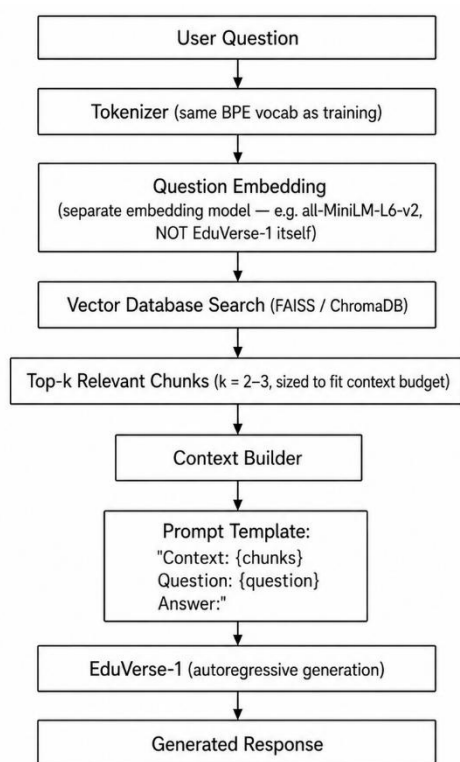
6. Optional Upgrades (if time allows)

These aren't required for a working prototype, but each is a well-known, well-documented improvement over the "vanilla" version above:

Upgrade	Replaces	Benefit
RoPE (Rotary Positional Embedding)	Learned positional encoding	Better length generalization, used by Llama/Qwen
RMSNorm	LayerNorm	Slightly cheaper, used by Llama/Qwen
SwiGLU FFN	GELU FFN	Small quality gain, used by Llama/Qwen

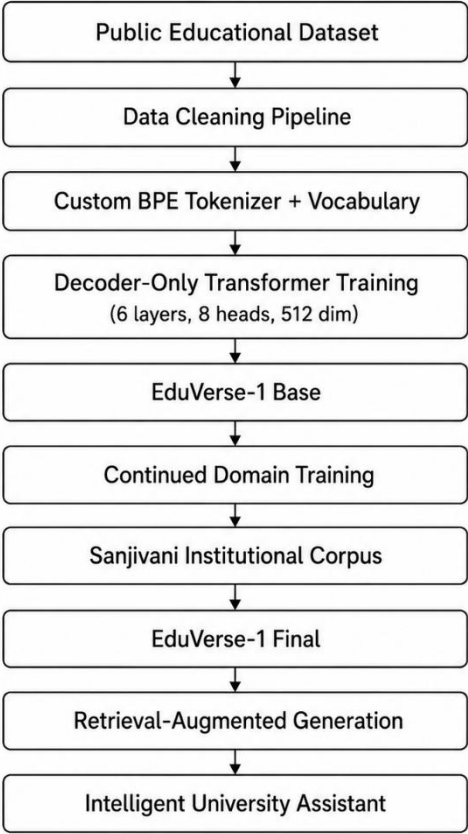
Start with the simple versions in §2–4 first get a working baseline before swapping in these.

7. Inference Pipeline (with RAG)



Note: with only 512 tokens of context, chunk size and top-k must be chosen carefully 2–3 short, high-relevance chunks will work far better than 5+ chunks that get truncated.

8. Summary Diagram



EDUVERSE-1 ARCHITECTURE

Domain-Specific Decoder-Only LLM with RAG for Intelligent University Assistant

